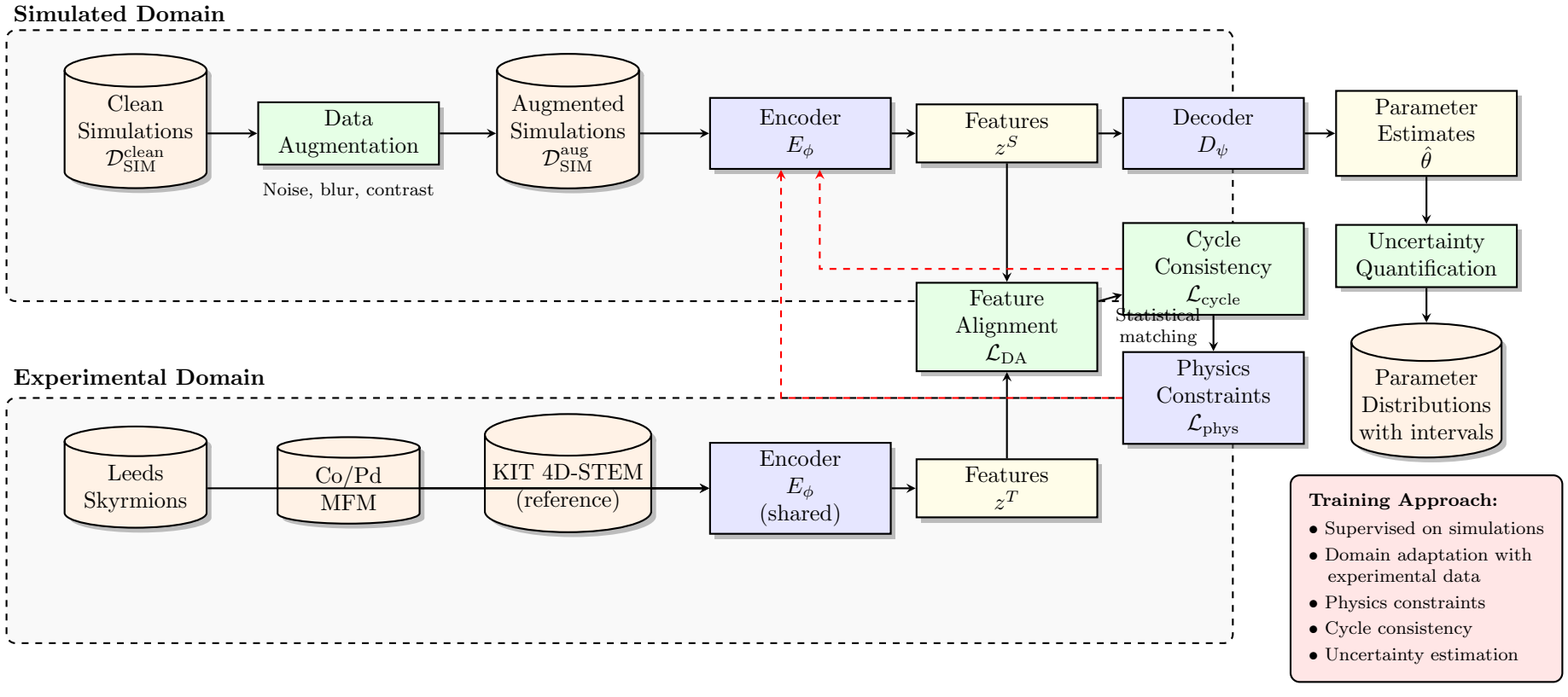

Simulated Domain
Clean Simulations $\mathcal{D}_{\text{SIM}}^{\text{clean}}$
Data Augmentation
Noise, blur, contrast
Augmented Simulations $\mathcal{D}_{\text{SIM}}^{\text{aug}}$
Encoder $E_\phi$
Features $z^S$
Decoder $D_\psi$
Parameter Estimates $\hat{\theta}$
Uncertainty Quantification
Parameter Distributions with intervals
Cycle Consistency $\mathcal{L}_{\text{cycle}}$
Feature Alignment $\mathcal{L}_{\text{DA}}$
Statistical matching
Physics Constraints $\mathcal{L}_{\text{phys}}$
Experimental Domain
Leeds Skyrmions
Co/Pd MFM
KIT 4D-STEM (reference)
Encoder $E_\phi$ (shared)
Features $z^T$
Training Approach:
• Supervised on simulations
• Domain adaptation with experimental data
• Physics constraints
• Cycle consistency
• Uncertainty estimation